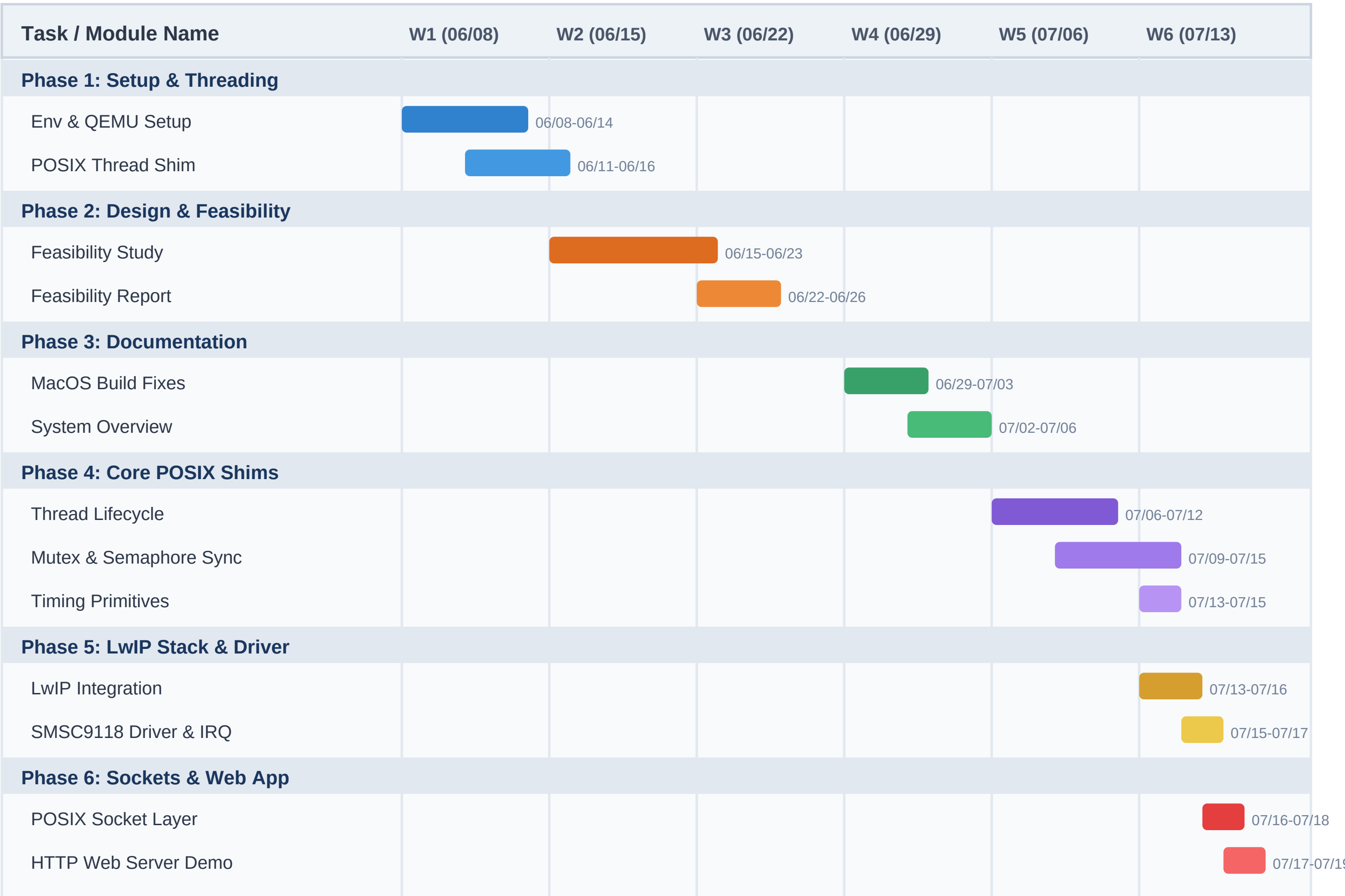
STM32 Simulated Linux: 6-Week Gantt Chart & Project Roadmap

Master schedule detailing the POSIX compatibility layer development on FreeRTOS in QEMU (June 8, 2026 – July 19, 2026).

1. Gantt Chart Timeline Diagram



2. Master Schedule Breakdown

ID	Phase	Full Task Name	Start	End	Days	Key Output / Deliverable
t1	Phase 1	**Initial QEMU & FreeRTOS Setup**	06/08	06/14	7	QEMU UART redirection in main.c
t2	Phase 1	**Basic POSIX Thread Translation Shim**	06/11	06/16	6	pthread_create mapping to xTaskCreate
t3	Phase 2	**Research & Feasibility Study**	06/15	06/23	9	Cortex-M3 RAM budget & POSIX evaluation
t4	Phase 2	**Feasibility Report & Heatmap**	06/22	06/26	5	Feasibility report & heatmap creation
t5	Phase 3	**MacOS Build Fixes & Debugging**	06/29	07/03	5	Toolchain build fix & Makefile updates
t6	Phase 3	**System Overview & Architecture**	07/02	07/06	5	Architectural guide SYSTEM_OVERVIEW.md
t7	Phase 4	**Thread Lifecycle (join, exit, detach)**	07/06	07/12	7	Thread registry & lifecycle in main_blinky.c
t8	Phase 4	**Mutex & Semaphore Synchronization**	07/09	07/15	7	pthread_mutex_t & counting sem_t

ID	Phase	Full Task Name	Start	End	Days	Key Output / Deliverable
t9	Phase 4	**Timing Primitives (sleep, usleep)**	07/13	07/15	3	Delay mapping to vTaskDelay ticks
t10	Phase 5	**LwIP TCP/IP Stack Integration**	07/13	07/16	4	LwIP OS layer adaptation in sys_arch.c
t11	Phase 5	**SMSC9118 Driver & NVIC IRQ**	07/15	07/17	3	Ethernet driver in ethernetif.c
t12	Phase 6	**POSIX Socket Shim Wrapper Layers**	07/16	07/18	3	BSD Socket APIs (socket, bind, listen)
t13	Phase 6	**HTTP Web Server Demo App**	07/17	07/19	3	Simulated POSIX Web Server on port 80/8080

3. Detailed Week-by-Week Breakdown

Week 1 (June 8 - June 14, 2026): Initial Bootstrap & Threading Shim

• Imported the FreeRTOS real-time kernel and MPS2 AN385 Cortex-M3 template configuration.

• Overrode stdout bindings in main.c to pipe console prints directly to QEMU UART0.

• Implemented initial pthread_create translation to FreeRTOS xTaskCreate task calls.

Week 2 (June 15 - June 21, 2026): Architectural Design & Research

• Analyzed memory constraints and RAM budget (70-80 KB limit) of the target board.

• Mapped priority scaling and stack depth allocations needed for simulating Linux threads.

• Formulated synchronization primitive strategy without requiring Unix host headers.

Week 3 (June 22 - June 28, 2026): Feasibility Report & Heatmap Creation

• Documented feasibility research results and API translation mappings.

• Created methodology evaluation heatmap scoring POSIX Shimming vs. Manual Rewriting.

• Prepared build instructions and repository setup in README.md.

Week 4 (June 29 - July 5, 2026): Tooling & Documentation Setup

• Resolved type-casting constraints inside pthread_create to fix toolchain build warnings.

• Updated build Makefile to ensure seamless cross-compilation across Linux and macOS.

• Authored SYSTEM_OVERVIEW.md detailing architectural layout and execution pathways.

Week 5 (July 6 - July 12, 2026): POSIX Shim Expansion

• Implemented thread lifecycle APIs: pthread_join, pthread_exit, pthread_detach, pthread_self.

• Mapped pthread_mutex_t to FreeRTOS mutexes and built custom sem_t counting semaphores.

• Integrated timing delay wrappers (sleep, usleep) mapped to FreeRTOS scheduler ticks.

Week 6 (July 13 - July 19, 2026): Networking Stack & Sockets Web Server

• Integrated LwIP TCP/IP stack with custom memory configuration tuned in lwipopts.h.

• Built sys_arch.c OS adaptation layer and SMSC9118 Ethernet driver in ethernetif.c.

• Mapped BSD socket APIs and built HTTP Web Server daemon listening on port 80/8080.